

Chapter 22

Linear Combinations; Point Estimation and Bias

Lecture 22 | Devore 9e: 5.5, 6.1

22.1 Linear combinations

Let $X_1, \dots, X_n$ have means $\mathbb{E}[X_i] = \mu_i$ and known covariances $\text{Cov}(X_i, X_j)$, and consider the statistic

$$Y = a_1 X_1 + a_2 X_2 + \dots + a_n X_n.$$

Theorem 22.1 (Mean).

$$\mathbb{E}[Y] = \sum_{i=1}^n a_i \mathbb{E}[X_i] = \sum_{i=1}^n a_i \mu_i.$$

This needs **no** independence assumption — expectation is always linear. Variance is different.

Theorem 22.2 (Variance).

$$\text{Var}(Y) = \text{Cov}(Y, Y) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j) = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} \begin{bmatrix} \text{Cov}(X_1, X_1) & \dots & \text{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \dots & \text{Cov}(X_n, X_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

If the X_i are independent, all off-diagonal terms vanish and

$$\text{Var}\left(\sum_i a_i X_i\right) = \sum_{i=1}^n a_i^2 \sigma_{X_i}^2$$

Remark 22.3. Note the a_i^2 : coefficients enter the mean linearly and the variance *quadratically*. In particular $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$ for independent X, Y — subtracting does not subtract variance, it adds it.

Example 22.4 (The hat check: how much does the count vary?). In Lecture 9 the number N of people, out of $n \geq 2$, who get their own hat back had $\mathbb{E}[N] = 1$. Find $\text{Var}(N)$.

As there, $N = \mathbf{1}_{A_1} + \cdots + \mathbf{1}_{A_n}$, where A_i is the event that person i gets their own hat. Now the dependence matters. Each indicator is Bernoulli with $p = \frac{1}{n}$, so $\text{Var}(\mathbf{1}_{A_i}) = \frac{1}{n}(1 - \frac{1}{n})$. For $i \neq j$ the product $\mathbf{1}_{A_i}\mathbf{1}_{A_j}$ indicates that both get their own hats, which has probability $\frac{1}{n} \cdot \frac{1}{n-1}$, so

$$\text{Cov}(\mathbf{1}_{A_i}, \mathbf{1}_{A_j}) = \frac{1}{n(n-1)} - \frac{1}{n^2} = \frac{1}{n^2(n-1)}.$$

There are n diagonal terms and $n(n-1)$ off-diagonal ones:

$$\text{Var}(N) = n \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right) + n(n-1) \cdot \frac{1}{n^2(n-1)} = 1 - \frac{1}{n} + \frac{1}{n} = 1.$$

Mean 1 and variance 1 for every n , just like a $\text{Poisson}(1)$ count — and N is in fact close to $\text{Poisson}(1)$: its chance of being 0 tends to e^{-1} , as in Lecture 4.

Example 22.5 (Diversification). Each of n assets has a return with variance σ^2 , and every pair has correlation ρ . An equal-weight portfolio holds $\frac{1}{n}$ of each. What is the variance of its return?

With every $a_i = \frac{1}{n}$, the n diagonal terms contribute $n \cdot \frac{\sigma^2}{n^2}$ and the $n(n-1)$ off-diagonal terms $n(n-1) \cdot \frac{\rho\sigma^2}{n^2}$:

$$\text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{\sigma^2}{n} + \frac{n-1}{n} \rho\sigma^2 = \sigma^2 \left(\rho + \frac{1-\rho}{n}\right) \xrightarrow{n \rightarrow \infty} \rho\sigma^2.$$

With $\rho = 0$ this is the σ^2/n of Lecture 21, and averaging removes all the risk. With $\rho = 0.3$ and $n = 10$ it is $0.37\sigma^2$, and no number of assets gets it below $0.3\sigma^2$: the shared part of the risk cannot be diversified away.

Example 22.6 (Mixing in a riskier investment). Two independent investments have returns X and Y with $\text{Var}(X) = 1$ and $\text{Var}(Y) = 4$. Put a fraction w into X and $1-w$ into Y . Which w gives the smallest variance?

By the boxed formula,

$$v(w) = \text{Var}(wX + (1-w)Y) = w^2 + 4(1-w)^2.$$

Setting $v'(w) = 2w - 8(1-w) = 0$ gives $w^* = 0.8$, and $v(0.8) = 0.64 + 0.16 = 0.8$. The mix is *less* variable than the safer investment alone: a little of an independent risky investment lowers the total variance, because its fluctuations partly cancel those of X . With covariance $c = \text{Cov}(X, Y)$

the same calculation gives $w^* = \frac{\sigma_Y^2 - c}{\sigma_X^2 + \sigma_Y^2 - 2c}$.

Theorem 22.7 (Normal is closed under linear combinations). *If $X_1, \dots, X_n$ are independent and each $X_i \sim \text{Normal}(\mu_i, \sigma_i^2)$, then*

$$\sum_{i=1}^n a_i X_i \sim \text{Normal}\left(\sum_i a_i \mu_i, \sum_i a_i^2 \sigma_i^2\right)$$

exactly — no approximation, no large- n requirement.

This is the fact behind “ $\bar{X} \sim \text{Normal}(\mu, \sigma^2/n)$ exactly for a normal sample” from last lecture.

22.2 Point estimation

Definition 22.8 (Point estimator and point estimate). Let θ be an unknown parameter of a distribution. A **point estimator** $\hat{\theta}$ of θ is a statistic — a random variable computed from $X_1, \dots, X_n$. The number it produces from actual data $x_1, \dots, x_n$ is the **point estimate**.

Remark 22.9. Same distinction as always: *estimator* = random variable (capital letters in); *estimate* = number (data in). An estimator has a distribution; an estimate does not.

Example 22.10 (The mean). $X_1, \dots, X_n$ i.i.d. with mean μ and variance σ^2 . Then $\hat{\mu} = \bar{X}$ is a point estimator of μ , and $\mathbb{E}[\bar{X}] = \mu$.

Example 22.11 (A proportion). Estimate the pass rate p of STAT400. Sample n students and let

$$X_i = \begin{cases} 1, & \text{student } i \text{ passes,} \\ 0, & \text{fails.} \end{cases}$$

Then $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$. Since $\mathbb{E}[X_i] = p$, we get $\mathbb{E}[\hat{p}] = p$.

Example 22.12 (A difference of means). iPhone battery life has mean μ_X ; Samsung's has mean μ_Y . To compare them, estimate $\mu_X - \mu_Y$ by

$$\bar{X} - \bar{Y} = \frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{m} \sum_{i=1}^m Y_i.$$

By the linear-combination rules, $\mathbb{E}[\bar{X} - \bar{Y}] = \mu_X - \mu_Y$ and (for independent samples) $\text{Var}(\bar{X} - \bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$.

Example 22.13 (Two estimators of the variance). For an i.i.d. sample, which of

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{or} \quad \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

should estimate σ^2 ? The next definition decides it.

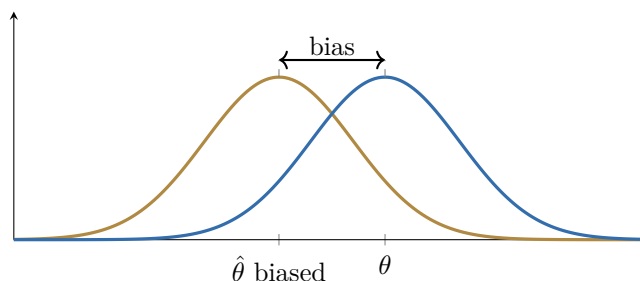
22.3 Bias

Definition 22.14 (Unbiased estimator). $\hat{\theta}$ is an **unbiased** estimator of θ if

$$\mathbb{E}[\hat{\theta}] = \theta.$$

In general $\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$.

Remark 22.15. Unbiased means: *on average over repeated samples*, the estimator hits the target. It says nothing about any single sample — an unbiased estimator can be wildly wrong every time, as long as the errors cancel. That is why the next lecture adds variance to the picture.



Theorem 22.16. For an i.i.d. sample with variance σ^2 ,

$$\mathbb{E} \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right] = \sigma^2,$$

so S^2 is unbiased, while $\frac{1}{n} \sum_i (X_i - \bar{X})^2$ has expectation $\frac{n-1}{n} \sigma^2$ and is biased downward.

Proof sketch (this is Homework). Use the identity $\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$, take expectations, and use $\mathbb{E}[(X_i - \mu)^2] = \sigma^2$ and $\text{Var}(\bar{X}) = \sigma^2/n$:

$$\mathbb{E} \left[\sum_i (X_i - \bar{X})^2 \right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2. \quad \square$$

22.4 A worked example: estimating the endpoint of a uniform

Example 22.17 (Apple Watch water resistance). The maximum time a watch survives under water is an unknown constant θ . You have n identical watches and test each one, obtaining $X_1, \dots, X_n$ i.i.d. Uniform $[0, \theta]$. Estimate θ .

The obvious estimator is

$$\hat{\theta} = \max(X_1, \dots, X_n),$$

and note immediately that $0 \leq \hat{\theta} \leq \theta$ always — it can never overshoot, which already smells biased. Let us compute.

For $0 \leq t \leq \theta$, using independence,

$$F_{\hat{\theta}}(t) = \mathbb{P}\{\max(X_1, \dots, X_n) \leq t\} = \mathbb{P}\{X_1 \leq t\} \cdots \mathbb{P}\{X_n \leq t\} = \left(\frac{t}{\theta}\right)^n,$$

so

$$f_{\hat{\theta}}(t) = \frac{n t^{n-1}}{\theta^n}, \quad 0 \leq t \leq \theta.$$

Therefore

$$\mathbb{E}[\hat{\theta}] = \int_0^\theta t \cdot \frac{n t^{n-1}}{\theta^n} dt = \frac{n}{\theta^n} \int_0^\theta t^n dt = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1} \theta \neq \theta.$$

So max is **biased**, with bias $-\theta/(n+1)$ — it systematically underestimates, exactly as the picture suggested. But the bias vanishes as $n \rightarrow \infty$, and the fix is immediate: rescale.

Theorem 22.18.

$$\hat{\theta}_{unbiased} = \frac{n+1}{n} \max(X_1, \dots, X_n)$$

is an unbiased estimator of θ .

Remark 22.19. This is the classic “German tank problem”: estimating the size of a population of serial numbers from the largest one observed. Notice that the naive estimator was not wrong so much as *systematically* wrong, and a systematic error is exactly the kind you can correct once you have computed it.